

Aryamaan Borgohain

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EDUCATION

Tezpur University

Integrated MSc. in Mathematics
CGPA/SGPA: 7.55/10 (1st Semester)

2025 – Present
Tezpur, India

Higher Secondary (AISSCE)

Passed AISSCE with 90.0%, Central Board of Secondary Education
Shrimanta Shankar Academy, Dispur, Guwahati

2025

Matriculation (AISSE)

Passed AISSE with 89.0%, Central Board of Secondary Education
Shrimanta Shankar Academy, Dispur, Guwahati

2023

MAJOR PROJECTS

OrbitForge: Celestial Mechanics Space-Flight Simulator

- Developed a high-accuracy orbital simulator in C++ and Python to solve Lambert's Boundary Value Problem.
- Implemented Encke's perturbations and higher-order symplectic integration to precisely calculate satellite and rocket launch windows to carry out interplanetary missions.

Numerical IVP Solvers & ODE Analysis Tools

- Built custom Python/Matplotlib engines to estimate Initial Value Problems (IVPs) and visualize direction fields for autonomous ODEs.
- Authored an independent (unpublished) paper analyzing Forward Euler, Picard's, Taylor series among other approximation methods for IVPs and Autonomous systems.

3D Surface Normal & Gradient Visualization Engine

- Computed partial derivatives and mapped gradient vector fields for explicitly defined 3D surfaces.
- Built a custom, multi-core frame-by-frame animation engine using Python, SciPy, NumPy, and OpenCV.

3D Mobile Skill Development Van Design

- Engineered a 3D mobile career development van.
- Included interior modeling and demo animations, of a modified vehicle chassis for a commercial agency.

GUI-based Educational Institute Management System

- Solo-developed a GUI-based educational institute management system with Python & MySQL integration, as my AISSCE 2025 Computer Science Project.
- Implemented Tkinter to provide a smooth interface with instant registration and batch system.

SKILLS

Languages : C++, C#, Python, LaTeX, HTML/CSS, SQL (MySQL)

Scientific Modules/Tools : SciPy, NumPy, SymPy, Matplotlib, OpenCV, Jamovi, Spyder IDE (Anaconda Distribution), Jupyter Notebook

Creative/Other Tools : Unity 3D, Blender, GitHub, DaVinci Resolve, OBS, Photoshop, Microsoft 365 Suite

COURSES/CERTIFICATES

Introduction to Python for Scientific Computing : University of Colorado Boulder (via Coursera)